



25th Anniversary of IFCWS - Indo-French Monsoon School on Remote-Sensing for Critical Zone Science 15th - 17th October 2026, Department of Civil Engineering, IISc

The Indo-French Cell for Water Science is celebrating its 25th anniversary this year. Understanding and monitoring the mechanisms leading to the large spatio-temporal variability of the water cycle and its impact on the carbon pools and fluxes has always been at the core of the research deployed by the IFCWS, and this research has benefited from the ever-increasing source of distributed data through Remote Sensing. Recently launched Earth Observation satellite missions, such as SWOT or NISAR, as well as missions scheduled for launch in the next few years, like TRISHNA, offer unprecedented opportunities to monitor key variables of the critical zone science such as vegetation cover, soil type, soil moisture, streamflow or evapotranspiration, with a high revisit capacity and for free.

In order to provide scientists and members of the New Space a vision and training on pathways to imagine future downstream services to a wide range of users, we have built together a Monsoon School in Remote-Sensing for Critical Zone Science with a two parts program: a one-day workshop which gathers specialists involved in the design of the satellite missions, which will present an overview of the state of the art of the current knowledge of the missions outline, benefits and expected precision, and a two-days training session focusing on mastering the retrieval methods of the relevant variables of the Critical Zone Science mentioned above, as well as the assimilation of the current and future remote sensing products and variables into a distributed modelling platform that runs from plot scale to region.

There is no registration fee. Outstation participants may be considered for a limited accommodation on campus, on request. To apply, send detailed bio-data (including date of birth, academic background, work experience) via e-mail to indo.french.monsoon.school@gmail.com. Please indicate in the e-mail i- the title of the Hands On you are interested in (check the feasibility due to parallel sessions) and ii- how the training fits and/or will be useful in your curriculum/academic project/research project. Attendees are expected to bring their own laptop.

Program

Day 1 (Thursday 15/10/2026): Workshop

Time	State of the Art Keynote Talks	Presenters
9:00-9:30	Introductory Remarks	
9:30-10:30	Microwave remote sensing for soil moisture monitoring	Mehrez Zribi (CESBIO)
10:30-11:30	Soil properties mapping : from Lab spectroscopy to remote sensing tools	Dharuma (NBSS) and Cécile Gomez (LISAH)
11:30-11:45	Coffee/Tea	
11:45-12:45	Remote sensing for carbon fluxes assessment	Debsunder Dutta (IISc)
12:45-1:00	OneWaterVision initiative: presentation	Sekhar Muddu (IISc)
1:00-1:45	Lunch	
1:45-2:45	Monitoring inland waters space - The SWOT Mission	Hind Oubanas (G-EAU)
2:45-3:45	Isolated Tree detection and monitoring with Pléiades and Venüs time series	Samuel Corgne (IFP)
3:45-4:00	Coffee/Tea	
4:00-5:00	Evapotranspiration from visible/NIR/TIR remote sensing	Eswar Rajasekaran (IITB)
5:00-6:00	Satellite Remote Sensing for Distributed Hydrological Modelling. Application to Water Resource Management	Vincent Simonneaux (CESBIO)

Day 2 (Friday 16/10/2026): Training 1st day

Time	Training Title	Presenters
9:30-1:00	Room 1: ET from TIR remote sensing part 1: introduction to the Energy Budget approach and presentation of the TRISHNA/SWEAT tool	Emmanuelle Sarrazin (CESBIO) and Eswar Rajasekaran (IITB)
	Room 2: Microwave remote sensing for soil moisture retrieval part 1	Nicolas Baghdadi and Hassan Bazzi (TETIS)
	Room 3: SWOT data for flood and river discharge characterization	J. Indu (IITB)
1:00-2:00	Lunch	
2:00-5:30	Room 1: ET from TIR remote sensing part 2: the TRISHNA/SWEAT tool	Emmanuelle Sarrazin and Kanishka Mallick (CESBIO)
	Room 2: Microwave remote sensing for soil moisture retrieval part 2	Nicolas Baghdadi and Hassan Bazzi (TETIS)
Break times	POSTER SESSIONS	Participating scientists & Scholars

Day 3 (Saturday 17/10/2026): Training 2nd day

Time	Training Title	Presenters
9:30-1:00	Room 1: MODSPA tools: Spatial Modeling with Remote Sensing, FAO-56, and Fine-Scale Water Balance for Irrigation Needs	Vincent Rivalland and Vincent Simonneaux (CESBIO)
	Room 2: lab spectroscopy for soil characterisation	Dharuma (NBSS) and Cécile Gomez (LISAH)
1:00-2:00	Lunch	
2:00-5:30	Room 1: Cropping pattern characterization with Venüs time series	Samuel Corgne (IFP)
	Room 2: Remote sensing for carbon fluxes assessment	Yuji Shigefuji (IISc) and Syam Chintala (IITH)
Break times	POSTER SESSIONS	Participating scientists & Scholars